

# HDOC Day-17

## *Menu-driven digit operations using switch-case and separate void functions*

### 1. Problem Statement

Write a menu-driven C program using switch-case. Each menu option must use a separate user-defined function with void return type and a simple function call. Each function reads its own input, processes it, and displays the result.

- Generate the largest possible number using its digits.
- Generate the smallest possible number using its digits.
- Swap the first and last digits.
- Check whether the number is a palindrome.

### 2. Algorithm

1. Start the program.
2. Display four menu choices and read the user's choice.
3. Use switch-case to call the corresponding void function.
4. Choice 1 – largestNumber(): read the number, count the frequency of digits 0–9, then build the result from digit 9 down to 0.
5. Choice 2 – smallestNumber(): read the number and count digit frequencies. Place the smallest non-zero digit first, then all zeros, then the remaining digits in ascending order.
6. Choice 3 – swapFirstLast(): read the number, obtain its first digit, last digit and middle part, then reconstruct the number after exchanging the first and last digits.
7. Choice 4 – checkPalindrome(): read the number, reverse its digits, compare the reversed number with the original, and display whether it is a palindrome.
8. For any other menu choice, display “Invalid choice”.
9. Stop.

### 3. Test Case Table

| Test | Choice / Operation  | Input | Expected Output | Purpose                                      |
|------|---------------------|-------|-----------------|----------------------------------------------|
| 1    | 1 – Largest         | 31042 | 43210           | Checks descending arrangement, including 0   |
| 2    | 2 – Smallest        | 31042 | 10234           | Checks that 0 is not placed at the beginning |
| 3    | 3 – Swap first/last | 12345 | 52341           | Checks correct exchange of end digits        |
| 4    | 4 – Palindrome      | 12321 | Palindrome      | Checks a palindrome number                   |
| 5    | 4 – Palindrome      | 12345 | Not palindrome  | Checks a non-palindrome number               |

### 4. C Program

```
#include <stdio.h>
```

```
void largestNumber(void);  
void smallestNumber(void);
```

```

void swapFirstLast(void);
void checkPalindrome(void);

int main(void) {
    int choice;
    printf("1. Generate largest possible number using its digits\n");
    printf("2. Generate smallest possible number using its digits\n");
    printf("3. Swap first and last digits\n");
    printf("4. Check palindrome\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);

    switch (choice) {
        case 1: largestNumber(); break;
        case 2: smallestNumber(); break;
        case 3: swapFirstLast(); break;
        case 4: checkPalindrome(); break;
        default: printf("Invalid choice\n");
    }
    return 0;
}

void largestNumber(void) {
    long long n, result = 0;
    int freq[10] = {0}, i;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    while (n > 0) { freq[n % 10]++; n /= 10; }
    for (i = 9; i >= 0; i--)
        while (freq[i]-- > 0) result = result * 10 + i;
    printf("Largest possible number = %lld\n", result);
}

void smallestNumber(void) {
    long long n, result = 0;
    int freq[10] = {0}, i;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    while (n > 0) { freq[n % 10]++; n /= 10; }
    for (i = 1; i <= 9; i++) {
        if (freq[i] > 0) {
            result = i; freq[i]--;
            while (freq[0]-- > 0) result *= 10;
            break;
        }
    }
    for (i = 0; i <= 9; i++)
        while (freq[i]-- > 0) result = result * 10 + i;
    printf("Smallest possible number = %lld\n", result);
}

void swapFirstLast(void) {
    long long n, temp, divisor = 1, middle, result;
    int first, last;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    if (n < 10) { printf("After swapping = %lld\n", n); return; }
    temp = n;
    while (temp >= 10) { temp /= 10; divisor *= 10; }
    first = n / divisor;
    last = n % 10;
    middle = (n % divisor) / 10;
    result = (long long)last * divisor + middle * 10 + first;
    printf("After swapping first and last digits = %lld\n", result);
}

void checkPalindrome(void) {
    long long n, temp, reverse = 0;
    printf("Enter a positive integer: ");
    scanf("%lld", &n);
    temp = n;
    while (temp > 0) { reverse = reverse * 10 + temp % 10; temp /= 10; }
    if (reverse == n) printf("The number is a palindrome.\n");
}

```

```
    else printf("The number is not a palindrome.\n");  
}
```

## 5. Sample Results

**Choice 1:** Input: 31042

Largest possible number = 43210

**Choice 2:** Input: 31042

Smallest possible number = 10234

**Choice 3:** Input: 12345

After swapping first and last digits = 52341

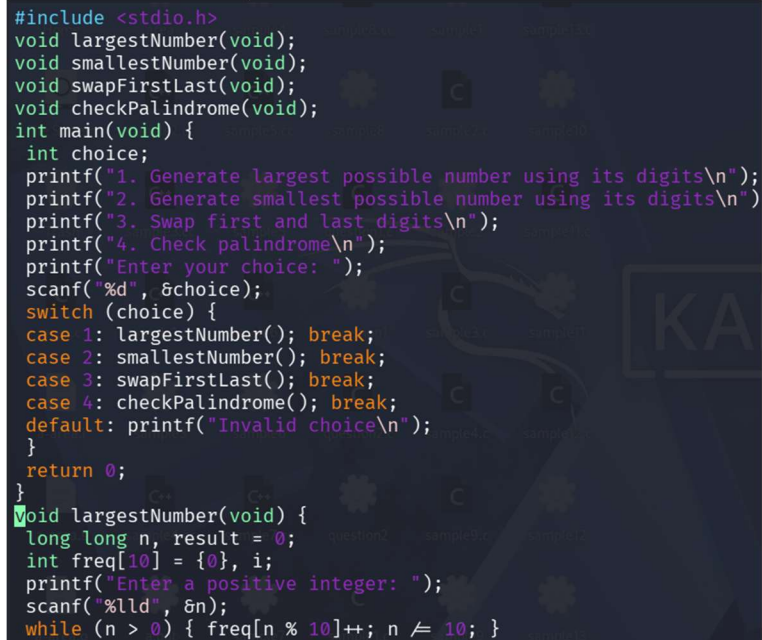
**Choice 4:** Input: 12321

The number is a palindrome.

## 6. Screenshot Space

Paste your own terminal screenshots after compiling and executing the program. Keep the menu choice and output visible in each screenshot.

### Screenshot 1 – Source code / successful compilation



```
#include <stdio.h>  
void largestNumber(void);  
void smallestNumber(void);  
void swapFirstLast(void);  
void checkPalindrome(void);  
int main(void) {  
    int choice;  
    printf("1. Generate largest possible number using its digits\n");  
    printf("2. Generate smallest possible number using its digits\n");  
    printf("3. Swap first and last digits\n");  
    printf("4. Check palindrome\n");  
    printf("Enter your choice: ");  
    scanf("%d", &choice);  
    switch (choice) {  
        case 1: largestNumber(); break;  
        case 2: smallestNumber(); break;  
        case 3: swapFirstLast(); break;  
        case 4: checkPalindrome(); break;  
        default: printf("Invalid choice\n");  
    }  
    return 0;  
}  
  
void largestNumber(void) {  
    long long n, result = 0;  
    int freq[10] = {0}, i;  
    printf("Enter a positive integer: ");  
    scanf("%lld", &n);  
    while (n > 0) { freq[n % 10]++; n /= 10; }
```

```

while (n > 0) { freq[n % 10]++; n /= 10; }
for (i = 9; i >= 0; i--)
while (freq[i]-- > 0) result = result * 10 + i;
printf("Largest possible number = %lld\n", result);
}

void smallestNumber(void) {
long long n, result = 0;
int freq[10] = {0}, i;
printf("Enter a positive integer: ");
scanf("%lld", &n);
while (n > 0) { freq[n % 10]++; n /= 10; }
for (i = 1; i <= 9; i++) {
if (freq[i] > 0) {
result = i; freq[i]--;
while (freq[0]-- > 0) result *= 10;
break;
}
}
for (i = 0; i <= 9; i++)
while (freq[i]-- > 0) result = result * 10 + i;
printf("Smallest possible number = %lld\n", result);
}

void swapFirstLast(void) {
long long n, temp, divisor = 1, middle, result;
int first, last;
printf("Enter a positive integer: ");
scanf("%lld", &n);
if (n < 10) { printf("After swapping = %lld\n", n); return; }

```

```

if (n < 10) { printf("After swapping = %lld\n", n); return; }
temp = n;
while (temp >= 10) { temp /= 10; divisor *= 10; }
first = n / divisor;
last = n % 10;
middle = (n / divisor) / 10;
result = (long long)last * divisor + middle * 10 + first;
printf("After swapping first and last digits = %lld\n", result);
}

void checkPalindrome(void) {
long long n, temp, reverse = 0;
printf("Enter a positive integer: ");
scanf("%lld", &n);
temp = n;
while (temp > 0) { reverse = reverse * 10 + temp % 10; temp /= 10; }
if (reverse == n) printf("The number is a palindrome.\n");
else printf("The number is not a palindrome.\n");
}

```

```

(kali@kali)-[~/Desktop]
$ vi sample13.c

(kali@kali)-[~/Desktop]
$ gcc sample13.c -o sample13 -lm

```

Screenshot 2 – Execution for Choice 1

```

(kali@kali)-[~/Desktop]
$ ./sample13
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check palindrome
Enter your choice: 1
Enter a positive integer: 31042
Largest possible number = 43210

```

Screenshot 3 – Execution for Choice 2

```

(kali@kali)-[~/Desktop]
$ ./sample13
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check palindrome
Enter your choice: 2
Enter a positive integer: 31042
Smallest possible number = 10234

```

Screenshot 4 – Execution for Choice 3

```
(kali㉿kali)-[~/Desktop]
$ ./sample13
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check palindrome
Enter your choice: 3
Enter a positive integer: 12345
After swapping first and last digits = 52341
```

## Screenshot 5 – Execution for Choice 4

```
(kali㉿kali)-[~/Desktop]
$ ./sample13
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter a positive integer: 12321
The number is a palindrome.
```

```
(kali㉿kali)-[~/Desktop]
$ ./sample13
1. Generate largest possible number using its digits
2. Generate smallest possible number using its digits
3. Swap first and last digits
4. Check palindrome
Enter your choice: 4
Enter a positive integer: 12345
The number is not a palindrome.
```